

MeetMind AI

Turn every conversation into actionable intelligence.

Why is the Earth cold in winter?

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Summary

["The Earth is closest to the Sun in early January (Perihelion) and farthest in early July (Aphelion).", "The distance difference is about 3 million miles out of 93 million miles.", "Seasons are caused by the Earth's tilt of 23.5 degrees, not its distance from the Sun.", "In winter, the northern hemisphere is tilted away from the Sun, while the southern hemisphere experiences summer."]

Action Items

No action items found.

Key Decisions

No key decisions found.

Open Questions

No open questions found.

Transcript

[00:00:00] Welcome to two-minute science videos.
[00:00:02] I'll bring you understand science and hundred-two.
[00:00:05] Why is the Earth cold and winter?
[00:00:07] Let's find out.
[00:00:07] If you ask someone on the street,
[00:00:09] why is it cold and winter?
[00:00:10] You're likely to get this response.
[00:00:12] Isn't the Earth farther from the Sun in the winter?
[00:00:15] This is a common misconception.
[00:00:16] The Earth is not farther from the Sun in the winter.
[00:00:18] In fact, the Earth is closest to the Sun in the winter.
[00:00:20] In early January, the Earth is actually nearest to the Sun.
[00:00:24] About 91 and a half million miles.
[00:00:26] This is called the Perihelion.
[00:00:27] Six months later in July,

[00:00:29] the Earth is far from the Sun.
[00:00:30] About 94.5 million miles.
[00:00:32] This is called the Aphelion.
[00:00:34] The difference is 3 million miles.
[00:00:36] What's 3 million miles among friends?
[00:00:38] And why is the Earth so cold in the winter?
[00:00:40] The seasons are actually due to Earth's tilt,
[00:00:42] not its distance from the Sun.
[00:00:43] The Earth is tilted about 23 and a half degrees relative to its orbital plane around the Sun.
[00:00:48] In some of the northern hemisphere is tilted towards the Sun.
[00:00:51] That's why it gets hot in the summer.
[00:00:52] In winter, the northern hemisphere is tilted away from the Sun.
[00:00:55] That's why it's cold in the winter.
[00:00:57] I use a globe to illustrate my point.
[00:00:59] This globe is tilted at 23 and a half degrees.
[00:01:01] Sun is over there.
[00:01:02] When it's winter in the northern hemisphere,
[00:01:04] the Sun shines more directly on the southern hemisphere.
[00:01:07] At summer in the southern hemisphere,
[00:01:09] when it's winter in the northern hemisphere.
[00:01:11] Six months later,
[00:01:12] after the Earth has gone around half way,
[00:01:14] the Sun shines more directly on the northern hemisphere.
[00:01:16] Therefore, it's summer in the northern hemisphere,
[00:01:18] but winter in the southern hemisphere.
[00:01:20] In summary, in early January,
[00:01:22] the Earth is actually nearest to the Sun.
[00:01:24] In early July, the Earth is farthest from the Sun.
[00:01:26] But the Earth is tilted by about 23 and a half degrees.
[00:01:30] That's why we have seasons.
[00:01:31] In the southern hemisphere, the seasons are opposite.
[00:01:33] The reason for the seasons is the Earth's tilt.
[00:01:36] To not the Earth's distance from the Sun.
[00:01:38] Three million miles out of 93 million miles.
[00:01:40] That's only minor, okay?
[00:01:42] So at ever season, it's just a go outside and enjoy it.
[00:01:45] Thank you for watching this two minutes science video.
[00:01:47] This is my YouTube channel for more videos.